

Glossary of Terms

COURSE 6 | DEVOPS ON AWS AND PROJECT MANAGEMENT

Project Management and Agile

Software development lifecycle (SDLC):

The stages that are completed while software is being built. The stages include planning and analysis, software design, implementation, and testing.

Waterfall: A software development process where software is built in a linear series of phases. Each phase must be complete before the process progresses to the next phase.

Agile: The larger umbrella of software development frameworks (including scrum and kanban) that are designed take an iterative approach to software development. Agile processes are more lightweight compared to the heavyweight processes of waterfall, which has strict planning and management.

(Agile, continued) Agile teams work with a focus on early and continuous delivery of software, accepting changing requirements, self-organizing teams, and shorter timeframes.

Scrum: An agile project management framework where software is built by smaller teams in short cycles called sprints. At the end of each sprint, the team delivers an incremental update to the software.

Product backlog: The one source of features that are completed and planned for a product.

Product owner: The representative of the stakeholders and the customers of a product. The product owner maintains the product backlog of features.

Daily scrum: A short meeting (up to a maximum of 15 minutes) where everyone in the team shares their progress on their assigned tasks for the sprint.

Sprint planning: A meeting to decide which items from the backlog the team will complete in the current sprint.

Sprint review: A meeting to inspect the work that was completed in the sprint. It also provides an opportunity for the team to give product demonstrations on new features.

Sprint retrospective: A meeting that looks back at the previous sprint, where team members discuss what went well and what areas need to improve.

Kanban: A visual process management framework where tasks are represented as cards that move through stages when they are completed. A kanban board gives all team members quick visibility into the efficiency and completeness of tasks.

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DevOps

Continuous integration and continuous delivery (CI/CD): A software development practice where automation is applied to the process of committing and testing code (integration) and the deployment of changes to different environments (delivery). The term can also mean continuous integration and continuous deployment if you remove all manual checks and automate the entire process, including the deployment of your code.

Source control (or version control): The practice of tracking and managing changes to code. Source control management (SCM) systems provide a running history of code development. They can help resolve conflicts when a team merges contributions from multiple sources.

Git: A distributed version control system that retains the full history of all changes in the code. Software development teams use Git as a single location for tracking changes from multiple engineers.

Jenkins: An open source automation server. Jenkins is used to automate building, testing, deploying, continuous integration, and continuous delivery of software.

Functional testing: Testing that is performed to see whether a software product functions as defined. Functional testing includes unit testing, regression testing, and integration testing.

Non-functional testing: Testing that verifies attributes outside of the defined functionality. Examples of non-functional testing include code quality testing, security testing, and performance testing.

Unit testing: Automated tests that are written to test a single unit of code in isolation. Mocking is frequently used to simulate other units of code that are used by the unit under test.

Regression testing: The process of rerunning previous tests to see whether new features created errors in previous functionality.

Integration testing: Testing to verify the interactions and integrations between multiple modules..

In place deployment: A form of application deployment where the application is replaced on existing infrastructure, such as Amazon Elastic Compute Cloud (Amazon EC2) instances.

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DevOps (Continued)

Rolling deployment: A form of application deployment where one of many instances is removed from production use so it can receive an application update. The updated instance is then returned to production use while the deployment completes the same process on all remaining instances.

Immutable deployment: A form of application deployment where updates are deployed to new resources, and the old resources are removed.

Blue-green deployment: A form of application deployment where network traffic is slowly shifted to newly deployed instances. After the deployment is complete, network traffic can be removed from old instances. The old instances can then be destroyed.

AWS Services

AWS CodeCommit: A fully managed source control service that hosts secure Git based repositories.

AWS CodeBuild: A fully managed continuous integration service that compiles source code, runs tests, and delivers software project artifacts.

AWS CodeDeploy: A deployment service that automates application deployments to EC2 instances, on-premises instances, serverless AWS Lambda functions, or Amazon Elastic Container Service (Amazon ECS) services.

AWS CodePipeline: A continuous delivery service that you can use to model, visualize, and automate the steps required to release software.

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